

Evote blockchain: Avoiding Coercion

Postdoctoral Research Proposal

Reynaldo C. Villena

¹reynaldocv @ gmail . com

Abstract. *This meta-paper describes the style to be used in articles and short papers for SBC conferences. For papers in English, you should add just an abstract while for the papers in Portuguese, we also ask for an abstract in Portuguese (“resumo”). In both cases, abstracts should not have more than 10 lines and must be in the first page of the paper.*

1. Introduction

Blockchain technology has emerged as a promising solution for secure and transparent digital transactions. Its decentralized nature and cryptographic principles make it an ideal candidate for applications that require trust and integrity. Blockchain allows to do transactions without the need for a trusted third party, as it relies on a distributed ledger that is maintained by a network of nodes. Each transaction records the details of the transaction (e.g., sender, receiver, amount, timestamp) and is recorded in a block, which is linked to the previous block, forming a chain of blocks. Moreover, blockchain technology provides anonymity and privacy, as users can interact with the system using pseudonymous addresses, without revealing their real identities. This feature is particularly important in applications that require confidentiality, such as voting systems, where the privacy of voters must be protected.

E-vote (electronic vote) is an process that allows voters to cast their votes electronically. One goal of this research proposal is to develop a blockchain-based e-voting system that ensures security, transparency, and privacy while avoiding coercion. The proposed system will leverage the decentralized nature of blockchain technology to provide a tamper-proof voting process. The e-voting process consider digital wallets for each voter and all candidates, where each wallet has a unique coin (e.g, in bitcoin, each voter’s wallet has a satoshi, which is the smallest unit of bitcoin). And, the voter can vote for a candidate by transferring its unique satoshi from their wallet to the candidate’s wallet. At the end of the voting process, count the number of satoshis in each candidate’s wallet to determine the winner.

One blockchain e-vote system records each transaction (vote) in a public ledger, which is accessible to all participants. And, one can infer wich digital wallet belongs to which candidate (At the end of the process, the quantity of satothis in each candidate’s wallet are related to the number of votes received by that candidate; therefore, it is possible to know which candidate owns that wallet). In resume; each transaction in blockchain e-vote reveals the voter’s choice (the candidate’s wallet that received the satoshi).

One property in e-vote is that the voter’s choice must be kept secret (i.e., the voter’s choice must not be revealed to anyone) and the voter receives a receipt that proves that the vote was cast (i.e., the voter can prove that they voted, but not for whom they voted)

and he can verify that their vote was counted correctly (i.e., the voter can check that their vote was included in the final tally

), but it is not possible to know which voter owns that wallet.

In the explained e-voting process, each transaction (vote) is recorded

All full papers and posters (short papers) submitted to some SBC conference, including any supporting documents, should be written in English or in Portuguese. The format paper should be A4 with single column, 3.5 cm for upper margin, 2.5 cm for bottom margin and 3.0 cm for lateral margins, without headers or footers. The main font must be Times, 12 point nominal size, with 6 points of space before each paragraph. Page numbers must be suppressed.

Full papers must respect the page limits defined by the conference. Conferences that publish just abstracts ask for **one**-page texts.

2. First Page

The first page must display the paper title, the name and address of the authors, the abstract in English and “resumo” in Portuguese (“resumos” are required only for papers written in Portuguese). The title must be centered over the whole page, in 16 point boldface font and with 12 points of space before itself. Author names must be centered in 12 point font, bold, all of them disposed in the same line, separated by commas and with 12 points of space after the title. Addresses must be centered in 12 point font, also with 12 points of space after the authors’ names. E-mail addresses should be written using font Courier New, 10 point nominal size, with 6 points of space before and 6 points of space after.

The abstract and “resumo” (if is the case) must be in 12 point Times font, indented 0.8cm on both sides. The word **Abstract** and **Resumo**, should be written in boldface and must precede the text.

3. CD-ROMs and Printed Proceedings

In some conferences, the papers are published on CD-ROM while only the abstract is published in the printed Proceedings. In this case, authors are invited to prepare two final versions of the paper. One, complete, to be published on the CD and the other, containing only the first page, with abstract and “resumo” (for papers in Portuguese).

4. Sections and Paragraphs

Section titles must be in boldface, 13pt, flush left. There should be an extra 12 pt of space before each title. Section numbering is optional. The first paragraph of each section should not be indented, while the first lines of subsequent paragraphs should be indented by 1.27 cm.

4.1. Subsections

The subsection titles must be in boldface, 12pt, flush left.

5. Figures and Captions

Figure and table captions should be centered if less than one line (Figure 1), otherwise justified and indented by 0.8cm on both margins, as shown in Figure 2. The caption font must be Helvetica, 10 point, boldface, with 6 points of space before and after each caption.



Figura 1. A typical figure

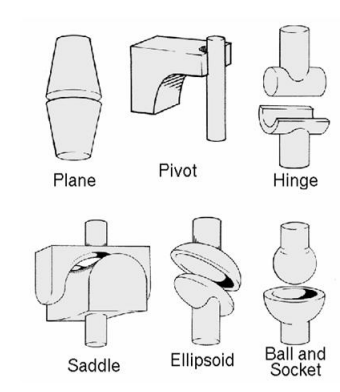


Figura 2. This figure is an example of a figure caption taking more than one line and justified considering margins mentioned in Section 5.

In tables, try to avoid the use of colored or shaded backgrounds, and avoid thick, doubled, or unnecessary framing lines. When reporting empirical data, do not use more decimal digits than warranted by their precision and reproducibility. Table caption must be placed before the table (see Table 1) and the font used must also be Helvetica, 10 point, boldface, with 6 points of space before and after each caption.

6. Images

All images and illustrations should be in black-and-white, or gray tones, excepting for the papers that will be electronically available (on CD-ROMs, internet, etc.). The image resolution on paper should be about 600 dpi for black-and-white images, and 150-300 dpi for grayscale images. Do not include images with excessive resolution, as they may take hours to print, without any visible difference in the result.

7. References

Bibliographic references must be unambiguous and uniform. We recommend giving the author names references in brackets, e.g. [Knuth 1984], [Boulic and Renault 1991], and [Smith and Jones 1999].

Tabela 1. Variables to be considered on the evaluation of interaction techniques

	Chessboard top view	Chessboard perspective view
Selection with side movements	6.02 ± 5.22	7.01±6.84
Selection with in- depth movements	6.29±4.99	12.22±11.33
Manipulation with side movements	4.66± 4.94	3.47±2.20
Manipulation with in- depth movements	5.71 ±4.55	5.37 ±3.28

The references must be listed using 12 point font size, with 6 points of space before each reference. The first line of each reference should not be indented, while the subsequent should be indented by 0.5 cm.

Referências

- Boulic, R. and Renault, O. (1991). 3d hierarchies for animation. In Magnenat-Thalmann, N. and Thalmann, D., editors, *New Trends in Animation and Visualization*. John Wiley & Sons Ltd.
- Knuth, D. E. (1984). *The T_EX Book*. Addison-Wesley, 15th edition.
- Smith, A. and Jones, B. (1999). On the complexity of computing. In Smith-Jones, A. B., editor, *Advances in Computer Science*, pages 555–566. Publishing Press.